TOPS Technologies

Interview Questions — Introduction to Data Analytics

Total Questions: 39

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1. What is data analytics, and why is it important for organizations today?

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2. Explain the difference between data, information, and insights using a simple example.

> **Hint:** Think about how raw numbers become meaningful and then actionable.

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3. What are the four main types of analytics, and what is the purpose of each?

> **Hint:** Recall: Descriptive, Diagnostic, Predictive, Prescriptive.

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4. What is the role of a Data Analyst (DA) compared to a Data Engineer (DE)?

> **Hint:** Focus on who prepares data and who analyzes it.

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5. Name at least three industry tools used for data analytics and mention one key use for each.

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6. If you have a sales dataset, how might it look different in Excel, SQL, Python, and Power BI?

> **Hint:** Think about the interface and capabilities of each tool.

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7. What is the difference between structured and unstructured data? Give an example of each.

> **Hint:** Consider tables vs. free-form text.

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8. List three common data formats you might encounter in analytics work.

> **Hint:** Think about files you open in Excel or import into Python.

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9. What is a relational database management system (RDBMS), and why is it used?

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10. Explain the difference between ETL and ELT processes in data handling.

> **Hint:** Focus on when transformation happens relative to loading data.

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11. How does OLTP differ from OLAP in terms of use case?

> **Hint:** Think about transactions vs. analysis.

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12. Why is domain knowledge important for a data analyst working on a retail project?

> **Hint:** Consider how business context affects which metrics matter.

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13. Give an example of a key performance indicator (KPI) for the finance domain.

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14. How would you define and use the metric 'Average Order Value' (AOV) in an analytics project?

> **Hint:** Think about what AOV tells you about customer behavior.

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15. If a client from the healthcare sector asks for a dashboard, what kind of business metrics might you include?

> **Hint:** Metrics could relate to patients, costs, or outcomes.

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16. What are the main steps in a typical data analytics workflow?

> **Hint:** Start from understanding the problem and end at sharing insights.

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17. Explain why data cleaning is a critical step before analysis.

> **Hint:** Consider what happens if there are errors or missing values.

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18. Describe the difference between data collection and data transformation in the analytics process.

> **Hint:** One is about gathering, the other about changing format or structure.

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19. How would you approach requirement gathering for a new analytics project?

> **Hint:** Think about communication with stakeholders.

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20. Imagine you have completed your analysis. What are some best practices for presenting insights to a non-technical audience?

> **Hint:** Consider the use of visuals and simple language.

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21. What is Exploratory Data Analysis (EDA), and why is it important at the start of a project?

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22. Describe the difference between univariate and bivariate analysis with examples.

> **Hint:** Think about analyzing one variable vs. two.

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23. How can you identify outliers in a dataset during EDA?

> **Hint:** Consider statistical methods or visual tools.

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24. If you are given a dataset in Excel, what is one simple way to spot trends or relationships between two columns?

> **Hint:** Think about charts or correlation functions.

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25. What is a multivariate analysis, and when would you use it?

> **Hint:** Think about analyzing more than two variables together.

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26. Suppose you notice an unexpected spike in sales in your EDA. What steps would you take to investigate this further?

> **Hint:** Think about checking data quality and external events.

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27. Why is it important to understand the flow of a dataset before starting your analysis?

> **Hint:** Consider how data moves from source to report.

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28. In a case study, how do you turn a business question into an analytics problem?

> **Hint:** Think about translating goals into measurable metrics.

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29. Imagine you are presented with a new case study dataset. What initial checks would you perform before analysis?

> **Hint:** Look for missing values, formats, and consistency.

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30. How would you use Excel to quickly summarize a dataset and spot any anomalies?

> **Hint:** Consider pivot tables or conditional formatting.

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31. How can you use SQL to filter data for only the last month's transactions in a sales table?

> **Hint:** Think about the WHERE clause and date functions.

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32. Give an example of how Python can be used for basic data cleaning.

> **Hint:** Consider removing duplicates or filling missing values.

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33. If you want to visualize sales by region in Power BI, what kind of chart would you choose and why?

> **Hint:** Consider which chart best shows comparisons across categories.

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34. Suppose your data is in JSON format. How would you load and work with this data in Python?

> **Hint:** Think about libraries or functions that handle JSON.

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35. How does understanding both business metrics and technical data help you deliver better analytics solutions?

> **Hint:** Consider how analysis is used for decision-making.

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36. If a dataset has both structured tables and unstructured text reviews, how might your approach to analysis change?

> **Hint:** Think about tools and methods suitable for each data type.

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37. You are asked to automate a monthly report that combines data from Excel and SQL. Briefly describe how you would approach this task.

> **Hint:** Consider data extraction, transformation, and automation tools.

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38. How could you use ChatGPT or another AI tool to help you understand a complex dataset or troubleshoot a data cleaning issue?

> **Hint:** Think about asking for code samples or explanations, but also how you would verify the AI's suggestions.

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39. What are some risks of relying solely on AI-generated insights or code when working on a data analytics project?

> **Hint:** Consider data privacy, accuracy, and the importance of human judgment.
